

RAJ FATEHVEER SINGH BRAR

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PROFILE SUMMARY

Computer Science undergraduate experienced in building automated data pipelines, decision-analysis engines, and business-facing reporting systems across banking and government internships, with measurable impact on processing time, reporting latency, and operational overhead. Strong Python/SQL foundation with hands-on exposure to AI/agent-based systems, cost-optimization modeling, and stakeholder-facing analytics. Eager to bring this cross-functional, data-driven rigor to AI cost governance, token attribution, and cross-platform usage optimization.

TECHNICAL SKILLS

AI Platforms & LLM Cost Ops (in progress): OpenAI, Anthropic, AWS Bedrock, LangChain, LangGraph, AI observability tools (LangSmith, Langfuse, Helicone), AWS Cost Explorer

Cost, Data & Financial Analytics: ETL Pipeline Design, BigQuery, BI Dashboards, Multi-Criteria Decision Analysis (TOPSIS), Budget/Resource Optimization (MILP)

Languages: Python, SQL, C++, JavaScript, R, HTML, CSS

AI/Agent Systems: Conversational/Agent-Based Systems, Intent & Entity Extraction, NLP (NLTK), Prompt-Driven Automation

Tools & Collaboration: Flask, Firebase, Git, Vercel/Render, SDLC, Cross-functional Stakeholder Management, Vendor & Requirements Gathering, Technical-to-Business Communication

WORK / INTERNSHIP EXPERIENCE

Business Intelligence Unit Intern | Axis Bank, Mumbai, India

May – July 2026

- Partnered with banking and analytics stakeholders to design and deploy Python-based ETL pipelines tracking daily demographic feeds, cutting manual review time and reducing operational cost overhead across cross-functional teams.
- Architected scalable pipeline logic to simplify loan-targeting workflows, improving data-processing throughput and enabling more granular, attribution-ready reporting.

Summer Intern | Ministry of Railways, New Delhi, India

June – July 2025

- Built an automated NLP pipeline in Python (NLTK) to structure unstructured passenger data, eliminating 20+ manual hours/week at 92% accuracy, directly reducing operational cost.
- Routed processed data into stakeholder-facing dashboards on BigQuery, cutting reporting latency from 3 days to 4 hours, mirroring the near-real-time usage/cost reporting this role requires.

PROJECTS

Quantitative Decision Analysis Package – Python, PyPI, MCDM/TOPSIS

- Built a Multiple-Criteria Decision Analysis engine implementing TOPSIS from scratch; published as an installable PyPI package and deployed as a Flask web service for weighted, ranked decision-making, directly transferable to cost/vendor trade-off evaluation.

Supply Chain Network Optimization – R, MILP

- Formulated a Mixed-Integer Linear Program to optimize network flow under production, demand, and budget constraints, directly applicable to cost-structure optimization and budget governance.

Conversational Agent – Rule-Based NLU Engine – Python, Finite-State Machine

- Built a fully local decision-making engine for a home robot using keyword-based entity extraction and a finite-state memory, deliberately engineered without any LLM API call, demonstrating cost-conscious AI system design and orchestration logic.

Data Visualization using Bokeh & Plotly – Python

- Built interactive, stakeholder-facing dashboards for exploratory and geospatial data, translating raw metrics into decision-ready visual reporting.

COURSES & CERTIFICATIONS

- CISCO – AICTE Virtual Internship (network security, cryptography, threat mitigation)
- Deloitte Australia Virtual Internship (digital transformation, incident response, enterprise risk management)

EDUCATION

Degree / Board	Institute	Year	CGPA / %
Bachelor of Engineering, CSE	Thapar Institute of Engineering and Technology, Patiala	2023 – 27	9.48 (up to 6th sem.)
CBSE XII	Bawa Lalvani Public School, Kapurthala	2023	96.66%
CBSE X	Bawa Lalvani Public School, Kapurthala	2021	99.80%

LANGUAGES KNOWN

- English (Fluent) • Hindi (Fluent) • Punjabi (Intermediate)